19 **D**

For first order diffraction, $d \sin\theta = \lambda$

$$\lambda / \mathrm{d} = \sin(28.6°)$$

For 2nd order maxima, $d \sin\theta_2 = 2\lambda$

$$\theta_2 = \sin^{-1} [2 \sin(28.6°)] = 73.2°$$

To find maximum n, $d \sin 90 = n_{max} \lambda$

$$n_{max} = d / \lambda = 2 \text{ (rounded down to nearest integer)}$$

Therefore, maximum number of intensity maxima observed is 5

20 **A**